

FORT WAYNE INTL, IN, USA

WMO: 725330

Lat: 40.971N

Lon: 85.206W

Elev: 241

StdP: 98.46

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14827

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.4	-15.3	-22.2	0.5	-18.0	-19.1	0.7	-14.8	13.3	-3.4	12.0	-3.2	4.4	260	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.6	23.5	31.2	22.8	29.7	22.1	25.3	30.6	24.3	28.9	23.5	27.8	4.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.1	28.4	22.9	18.1	27.3	22.0	17.2	26.1	78.6	30.4	74.6	29.1	70.9	27.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.5	10.1	8.8	-21.7	34.6	4.0	2.0	-24.6	36.0	-26.9	37.2	-29.1	38.3	-32.0	39.7		
			-22.0	26.7	3.8	1.1	-24.7	27.5	-27.0	28.2	-29.1	28.8	-31.9	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-3.8	-1.8	3.7	10.2	16.3	21.6	23.1	22.0	18.4	12.0	5.1	-0.7
	DBStd	10.67	6.82	6.29	5.99	5.08	4.71	3.67	3.02	2.80	3.95	4.87	5.21	5.82
	HDD10.0	1568	429	333	212	57	5	0	0	0	1	35	162	334
	HDD18.3	3304	686	564	455	247	97	13	2	4	47	206	396	589
	CDD10.0	1779	2	2	16	64	200	347	407	372	253	96	17	3
	CDD18.3	474	0	0	1	4	34	109	151	118	49	8	0	0
	CDH23.3	4159	0	0	8	48	338	951	1361	928	449	77	1	0
	CDH26.7	1344	0	0	1	6	99	338	489	263	133	15	0	0
Wind	WSAvg	4.1	5.0	4.8	4.7	4.9	4.2	3.6	3.0	2.7	3.0	3.9	4.4	4.6
Precipitation	PrecAvg	939	55	51	71	89	102	106	94	90	72	68	73	68
	PrecMax	1385	138	174	145	180	259	304	279	246	172	227	204	192
	PrecMin	620	10	8	12	26	26	17	10	11	9	4	15	11
	PrecStd	162	32	30	29	38	51	53	52	46	39	41	37	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	17.1	24.1	27.6	31.8	34.2	34.7	32.8	32.6	28.7	21.4	16.3
		MCWB	13.1	12.8	16.8	18.1	20.5	23.5	24.7	24.4	22.3	20.2	14.7	14.4
	2%	DB	11.7	13.2	19.6	24.5	29.3	32.2	32.8	31.0	30.3	25.5	18.0	13.1
		MCWB	10.2	10.7	13.7	16.3	20.0	22.7	24.4	23.8	21.5	17.8	13.8	11.4
	5%	DB	8.6	9.6	16.1	22.1	27.2	30.4	31.1	29.5	27.9	22.7	15.8	10.1
		MCWB	7.2	7.3	12.0	14.9	19.3	22.0	23.3	22.8	20.3	16.9	12.6	8.4
	10%	DB	5.0	6.2	13.2	19.3	24.8	28.5	29.5	28.0	26.0	20.3	13.4	7.0
		MCWB	3.6	4.4	10.0	13.4	18.2	21.2	22.5	21.9	19.3	15.6	10.5	5.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.2	14.3	17.6	20.0	23.0	25.4	26.8	26.2	24.6	21.8	16.8	14.5
		MCDB	14.4	15.8	23.7	25.6	28.2	31.6	32.2	31.1	29.9	26.7	19.7	16.3
	2%	WB	10.7	10.9	14.9	17.7	21.7	24.2	25.6	25.1	23.0	19.7	15.0	11.5
		MCDB	11.5	12.4	18.5	22.5	27.1	29.7	30.8	29.5	27.6	23.3	17.0	12.7
	5%	WB	7.5	7.4	12.7	15.9	20.5	23.2	24.6	24.0	21.8	18.0	13.0	8.2
		MCDB	8.8	9.6	15.7	20.2	25.2	28.4	29.3	27.6	25.6	21.5	15.5	9.7
	10%	WB	3.7	4.6	10.1	14.2	19.2	22.3	23.6	23.0	20.6	16.2	10.7	5.5
		MCDB	4.8	6.3	13.2	18.8	23.5	26.9	27.9	26.5	24.2	19.6	13.0	6.9
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.2	9.8	11.6	11.4	10.8	10.9	10.8	12.0	10.9	8.8	7.1
		MCDBR	9.6	11.6	14.2	15.2	14.2	13.0	12.2	11.9	14.3	14.1	12.1	10.1
		MCWBR	8.6	9.2	9.7	8.9	7.2	5.6	5.3	5.8	6.7	7.9	8.7	8.7
	5% WB	MCDBR	9.5	11.1	13.1	13.2	11.9	11.0	10.7	10.2	11.4	11.8	10.7	9.4
		MCWBR	8.8	9.5	9.6	8.9	6.9	5.5	5.3	5.6	6.1	7.6	8.6	8.5
Clear-Sky Solar Irradiance	taub	0.304	0.333	0.350	0.408	0.453	0.445	0.446	0.447	0.410	0.356	0.323	0.302	
	taud	2.390	2.296	2.345	2.191	2.109	2.184	2.198	2.179	2.255	2.401	2.448	2.433	
	Ebn at Noon	849	870	898	863	829	835	830	817	823	836	822	822	
	Edh at Noon	82	105	112	140	156	145	142	141	122	94	78	73	
All-Sky Solar Radiation	RadAvg	1.58	2.48	3.54	4.66	5.38	6.04	6.08	5.38	4.39	2.95	1.90	1.33	
	RadStd	0.16	0.30	0.30	0.42	0.49	0.42	0.33	0.28	0.36	0.27	0.25	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page